

Wearable device requirements

Project: AmazeLight

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Preface

This document was generated with the wearable device specification tool, that enables a researcher to gather important requirements for a wearable device in a study, so that the device can be selected based on the requirements.

Restore this configuration in the tool through this link

In some situations, a device might not be able to meet all the requirements, and the researcher might need to make a decision on which requirements are most important. This document is a resource to guide the decision process between researchers and manufacturers.

General information

Type(s) of study: Observational, Cross-sectional

This project requires 50 devices

Description of the (main) use case:

Participants are fitted with a device at the study visit and instructed to wear it continuously for 14 days, removing it only for intense water-based activities. Data are wirelessly uploaded to a secure cloud server once a day. Participants complete one online questionnaire for chronotype. Primary outcomes are 24-h melanopic dose, timing of first >250 lux epoch, and their correlation with midpoint of sleep on free days (MSFsc).

Hardware requirements

Aspect	Requirement	Device specs
Mounting	Necklace	
Max dimensions (WxBxH)	3 x 5 x 1.5 cm	
Battery: Continuous runtime, based on the specified recording/sampling interval	≥2 weeks	

Data requirements

Aspect	Requirement	Device specs
Minimum recording interval	≤ 60 seconds	
Flexibility of recording interval	Recording interval needs to be freely adjustable	

Other requirements

Aspect	Requirement	Device specs
Available budget per device	500 €	